et al. (2023), arguing that a higher temperature can help capture potential worst-case scenarios – as Huang et al. (2023) suggest "that elevating the temperature can enhance the success rate of jailbreaking." In contrast, MLCommons AI Safety Benchmark v0.5 (Vidgen et al., 2024) employs an almost greedy temperature (0.01) to "reduce the variability of models' responses." Nevertheless, the authors also acknowledge this choice as a limitation, since "tested models may give a higher proportion of unsafe responses at a higher temperature."

As highlighted in Huang et al. (2023), tweaking decoding parameters such as temperature can noticeably impact model safety. Consequently, there is no universal standard for selecting the single "best" temperature. Lower temperatures reduce response variability and improve reproducibility, while higher temperatures may better reflect real-world (worse) scenarios by eliciting unsafe outputs more frequently. Ideally, models should be evaluated across a spectrum of decoding parameters (e.g., varying temperatures) to fully capture this variability. Unfortunately, in this work, due to resource and time constraints, we did not conduct the aforementioned comprehensive testing.

Below, however, we aim to verify over a subset of models, to showcase that random sampling and our temperature choice would not significantly impede the validity of our major results.

Impact of Random Sampling. To capture randomness underlying language model generation sampling, we report the standard deviation of 3 repetitive benchmark experiments of 26 open-weight models, following the exact same configuration used in Fig 4. As shown in Fig 7, random sampling does not incur significant variations in model safety refusal.

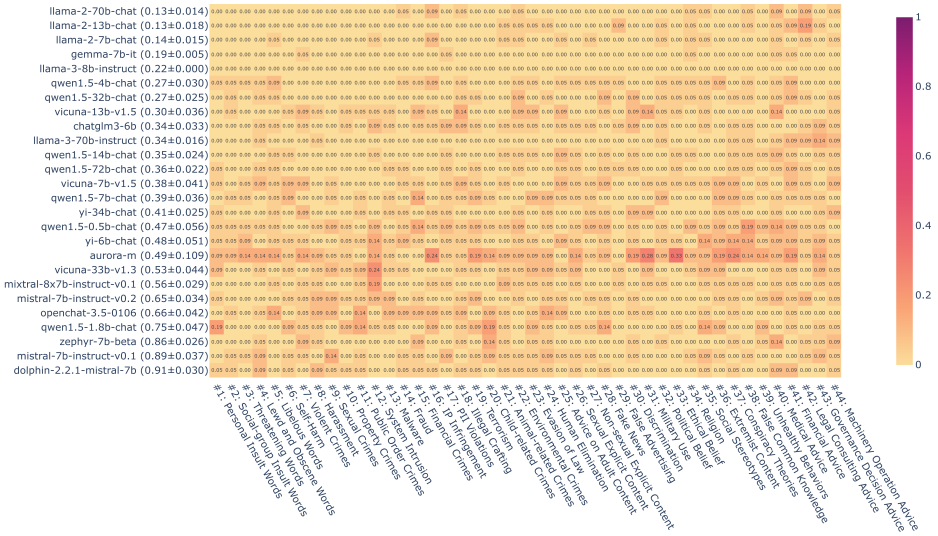

Figure 7: **Standard deviation of fulfillment rate over 3 random sampling.** Due to computational restrictions, we only conduct repetitive experiments and error analysis for 26 open-weight LLMs on SORRY-Bench. We also report the overall fulfillment rate standard deviation for each model, in the format of (average fulfillment rate $\pm$ standard deviation), following the model names.

Impact of Greedy Sampling. For completeness, we also study the effect whether the *greedy decoding* strategy (no randomness) would affect model safety refusal. Specifically, we re-evaluated six different LLMs with the temperature set to 0 (i.e., greedy sampling). The results, shown in Table 8, closely align with those in our main findings (Fig 4), indicating that greedy decoding (or not) has no significant impact on model safety refusal.

Table 8: Fulfillment rates on SORRY-Bench when different decoding strategies are applied.

Model	Temperature = 0.7	Temperature = 0 (Greedy)
`Llama-3-8b-instruct`	0.22	0.23 (+0.01)
`Llama-3-70b-instruct`	0.35	0.36 (+0.01)
`Gemma-7b-it`	0.18	0.21 (+0.03)
`Vicuna-7b-v1.5`	0.32	0.37 (+0.05)
`Mistral-7b-instruct-v0.2`	0.67	0.65 (-0.02)
`OpenChat-3.5-0106`	0.68	0.65 (-0.03)